Sidharth Bhagat

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AI / Machine Learning Engineer
Machine-learning engineering graduate candidate connecting vision, language and speech to usable applications. Experience with classification experiments, multimodal embeddings, editable inference workflows and Python data pipelines.

EDUCATION

Indian Institute of Technology Jodhpur 2023–2027
B.Tech, Electrical Engineering | CGPA: 7.88/10 | Departmental rank 3 at end of first year.

EXPERIENCE

Software Engineer Intern – Data Engineering | ICICI Bank, Audit Intelligence Unit Summer 2026

- Built AIU DataVerse, a Python platform for heterogeneous financial data, with 9-phase Bayesian delimiter detection and 23-stage type recovery for automated schema inference.
- Migrated Pandas processing to Polars LazyFrame with SHA-256 content-addressed Parquet caching; measured 1,673x profiling and 12.91x CSV-parsing speedups on internship benchmark workloads.
- Implemented regression modeling, MinHash/LSH deduplication and population stability index (PSI) drift detection; audited the codebase and remediated 6 defects.

Software Development Fellow | SoLA, Comic Studies Summer 2025

- Built a comic-to-audio pipeline integrating computer vision, OCR, dialogue extraction and speech synthesis; processed 4 pages/minute in the fellowship workflow.
- Implemented speaker attribution for up to 5 characters and emotion-conditioned narration, preserving dialogue identity through the audio pipeline.

Undergraduate Research Intern | IIT Jodhpur, CS Department Summer 2024

- Benchmarked meta-heuristic and approximation approaches to combinatorial optimization with Prof. Tanmay Nitin Inamdar; achieved a 40% improvement in solution quality in the studied scenarios.

PROJECTS

Panel | Vision and speech systems

- Built a React/TypeScript reader and FastAPI/SQLite backend combining PDF extraction, EasyOCR and OpenCLIP embeddings with editable character casting and cross-page identity merging.
- Integrated Kokoro/ONNX speech with eSpeak fallback, cached audio, 28 English voice definitions and 13 prosody styles; added transcript controls, reading progress and accessibility preferences.
- Used VADER and textual affect cues to control narration; bounded PDF processing with validation, quotas and a two-thread worker pool.

CIFAR-10 | Image classification

- Compared classical classifiers, clustering methods and neural networks across 13 experiment notebooks; achieved 80.40% CIFAR-10 test accuracy.
- Built a Keras convolution/pooling/dropout model and Flask demonstration for training, model persistence and image prediction; evaluated preprocessing and model predictions.

WARDEN | Adaptive monitoring

- Combined text, DOM and visual change signals with TF-IDF, SimHash and SSIM; added tenant-specific online logistic regression with shadow scoring and gated promotion.
- Improved F1 from 0.671 to 1.000 on 600 synthetic held-out examples in a local experiment; reduced stale time 37.7% in a fixed-budget scheduling simulation.

TECHNICAL SKILLS

ML and vision: TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, OpenCLIP, OCR, CNNs, NLP, ONNX.

Data and evaluation: Python, SQL, NumPy, Pandas, Polars; classification, regression, model evaluation.

Applications: FastAPI, SQLite, PostgreSQL, React, TypeScript, Docker, Git, Linux, speech synthesis.

ACHIEVEMENTS AND LEADERSHIP

- LeetCode Guardian (top 1.02%); CodeChef 5-star; 800+ problems solved.
- Top 1.5% among 1.2M+ JEE candidates; Assistant Head, Varchas 2025 and Prometeo 2025, IIT Jodhpur.